

AI SYSTEM DESIGN



PROBLEM STATEMENT



Timing Rigidity

Standard reminders use fixed "minutes before" logic, failing to account for the actual significance of the event.



Priority Blindness

Current systems treat a casual sync and a critical viva equally, leading to preventable misses or notification fatigue.



Lack of Autonomy

Existing tools require manual threshold setting, lacking the "intelligence" to sense, analyze, and act

INTELLIGENT AGENT CLASSIFICATION

Utility-Based Agent

The system is categorized as a **Utility-Based Agent** because it makes decisions based on a performance measure (utility) that maps a state to a real number

Reasoning & Logic

- ✓ **Goal vs Utility:** Beyond just reaching the 'notified' state, it seeks the *optimal* notification time.
- ✓ **State Analysis:** The system evaluates the current state to determine the best possible action.

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| SYSTEM ARCHITECTURE

Top 8 Tech Stack Examples

Full-Stack Integration

Frontend (React): Responsive dashboard for user event entry and monitoring.

Backend (Node/Express): Middleware handling API requests and data

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Frontend (React): Responsive dashboard for user event entry and monitoring.

Backend (Node/Express): Middleware handling API requests and data flow.

Database (MongoDB): Stores event metadata, priority levels, and

Top 8 Tec

LAMP Stack

OPERATIONAL WORK FLOW



AGENT LOGIC: UTILITY CALCULATIONS

$$U_{\text{alert}} = \begin{cases} 1 & \text{if } (T_{\text{event}} - T_{\text{now}}) \leq \text{Threshold}_{\text{Priority}} \\ 0 & \text{otherwise} \end{cases}$$

PRIORITY LEVEL	UTILITY THRESHOLD	DECISION ACTION
HIGH	20 Minutes	Immediate preparation alert
MEDIUM	10 Minutes	Standard reminder alert
LOW	5 Minutes	Standard reminder alert

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UTILITY THRESHOLD

DECISION ACTION

PRIORITY LEVEL

REAL-WORLD USE CASES

Enterprise Scheduling

Automating prep-times for high-stakes meetings, allowing executives focus-time before critical syncs.

Health & Medication

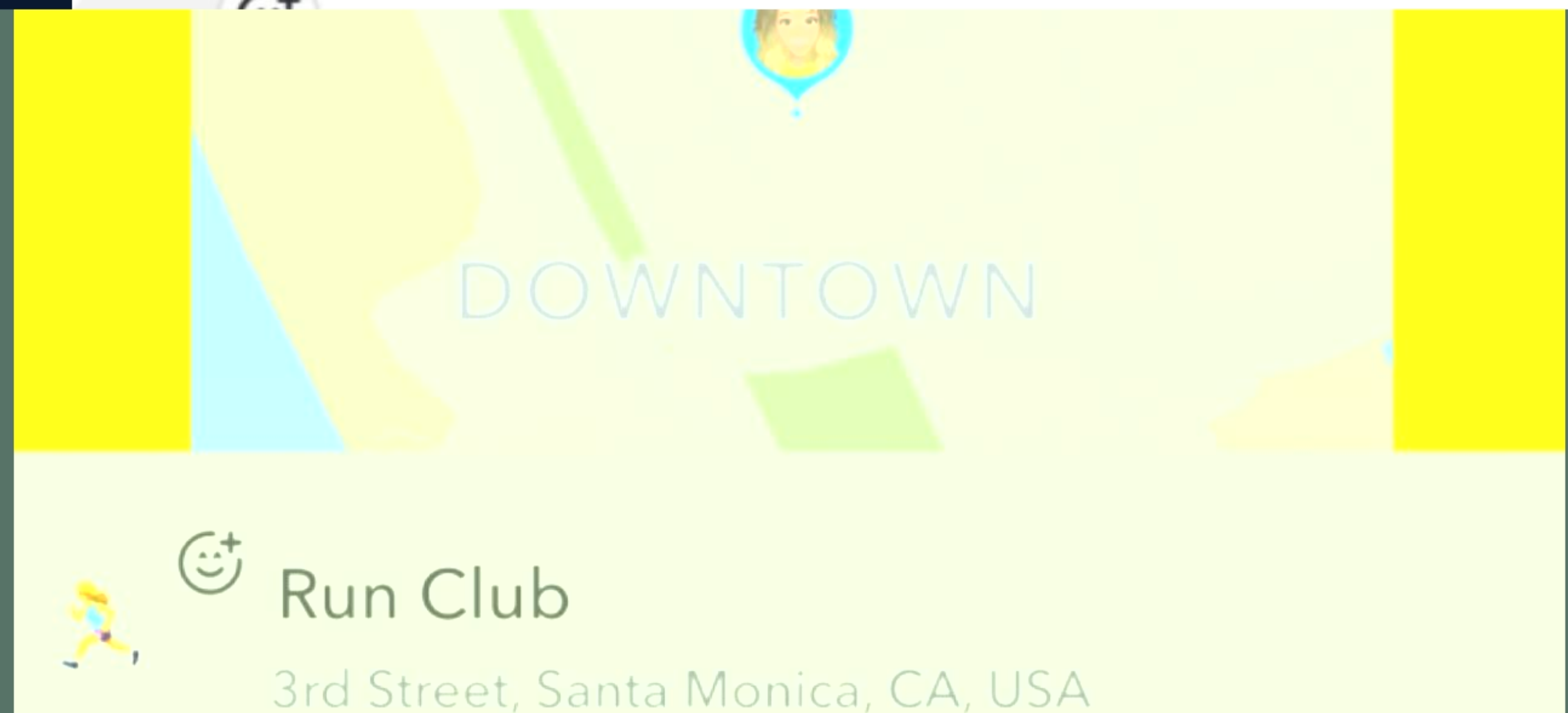
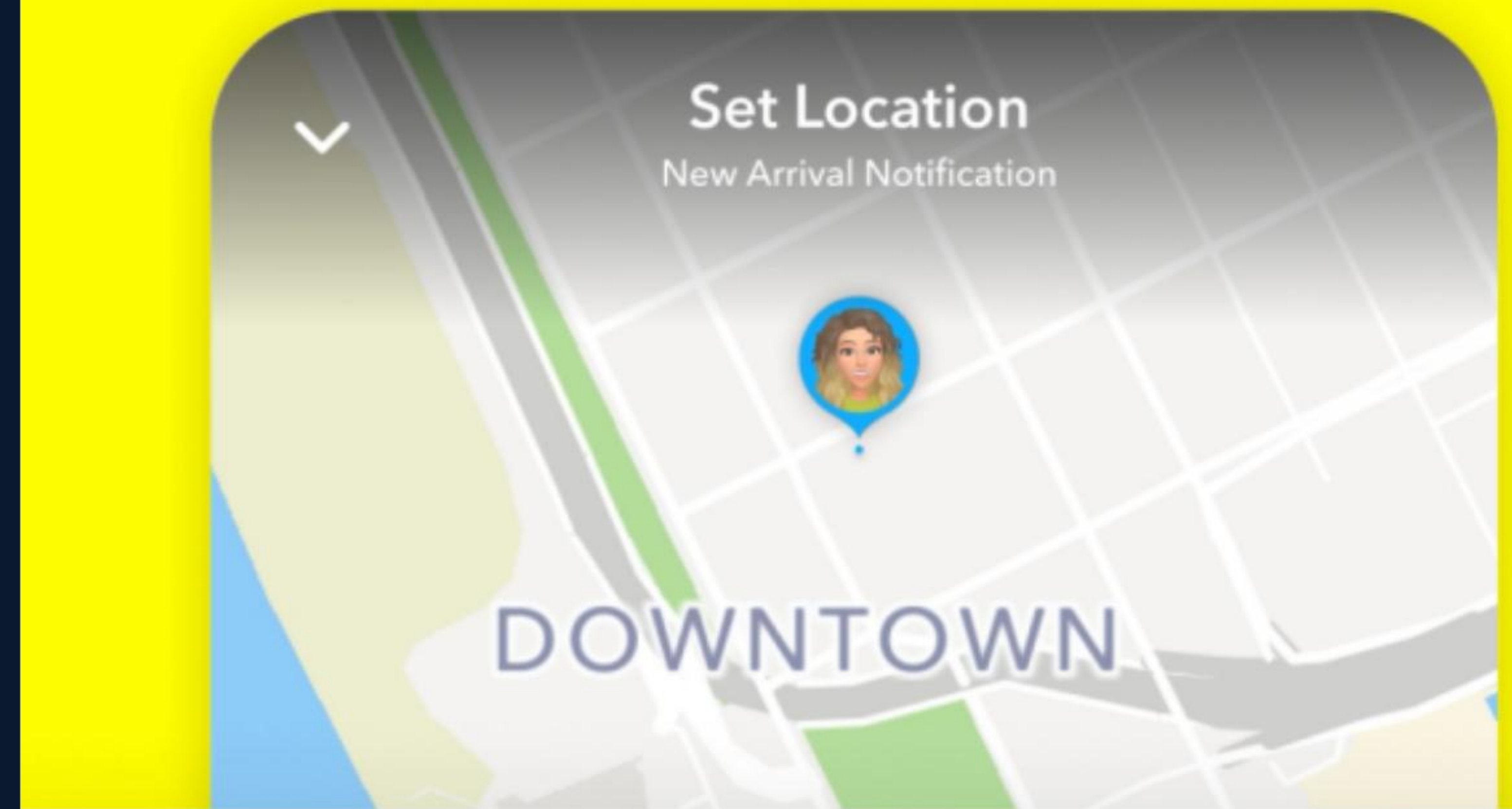
Adaptive medication reminders that prioritize critical doses over standard wellness check-ins.

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Conclusion

The Smart Autonomous Event Reminder Agent successfully implements a utility-based decision model to bridge the gap between simple scheduling and intelligent preparedness.

Autonomy: Independent sense-act cycle.

Efficiency: Optimized notification density.

Scalability: Full-stack ready deployment.

THANK YOU FOR YOUR ATTENTION!

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Autonomy: Independent

Efficiency: Optimized

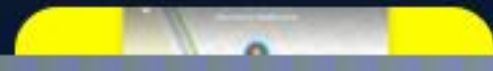
Scalability: Full-stack

IMAGE SOURCES



<https://www.mindpathtech.com/wp-content/uploads/2025/07/Top-8-Tech-Stack-Examples--scaled.jpg>

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